

Project ID: BT0678

Title	Explainable AI for clinical Decision Support in Heart Disease Prediction		
Project Type	☐ Community based design problem (Interdisciplinary) ☐ App Development / Utility ☐ IOT/Hardware based ☒ AI/ML/Data Science ☒ Healthcare Projects	Project Outcome	☒ Project and Research Paper ☐ Project and Patent ☐ Project and Book Chapter
		SDG Goal Number	3
Publication Target	☒ SCOPUS Journal ☐ SCOPUS Conference ☐ SCOPUS Book Chapter ☐ SCI Journal	Guide Name: Dr. Santosh Kumar	

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Sustainable Development Goals (SDG Goals), Targets and Indicators

- Goal 1: End poverty in all its forms everywhere
 Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture
 Goal 3: Ensure healthy lives and promote well-being for all at all ages
 Goal 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
 Goal 5: Achieve gender equality and empower all women and girls
 Goal 6: Ensure availability and sustainable management of water and sanitation for all
 Goal 7: Ensure access to affordable, reliable, sustainable and modern energy for all
 Goal 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
 Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
 Goal 10: Reduce inequality within and among countries
 Goal 11: Make cities and human settlements inclusive, safe, resilient and sustainable
 Goal 12: Ensure sustainable consumption and production patterns
 Goal 13: Take urgent action to combat climate change and its impacts
 Goal 14: Conserve and sustainably use the oceans, seas and marine resources for sustainable development
 Goal 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
 Goal 16: Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
 Goal 17: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Explainable AI for Clinical Decision Support in Heart Disease Prediction

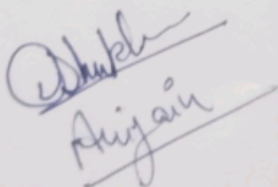
Domain: AI/ML

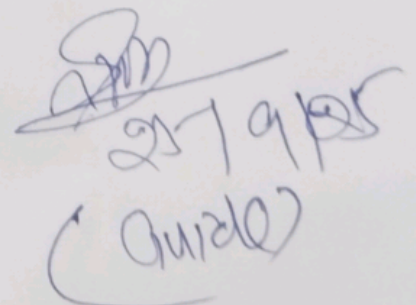
Abstract:

The majority of countries globally find that cardiovascular diseases have the greatest death rate. Early detection of heart disease is necessary to prevent deaths from it. The World Health Organization (WHO) noted that in 2019, 32% of all fatalities were attributed to heart-related issues. In India, the Ministry of Health and Family Welfare reported that cardiovascular diseases accounted for 28.1% of the overall deaths. Heart disease remains a major global health concern, particularly in regions with limited access to diagnostic facilities. Machine learning has the potential to significantly enhance the accuracy, efficiency, and speed of heart disease diagnosis. With the increasing availability of structured and unstructured data and the swift progress of analytical techniques, Artificial Intelligence (AI) is bringing a revolution to the healthcare industry. With the increasingly indispensable role of AI in healthcare, there are growing concerns over the lack of transparency and explainability. This lack of interpretability reduces trust among clinicians and patients, posing a significant barrier to deployment. This is where Explainable Artificial Intelligence (XAI) comes into the picture. This study proposes an Explainable AI driven clinical decision support system for heart disease prediction. Using publicly available datasets, various ML classifiers such as Logistic Regression, Random Forest, and XGBoost are implemented to predict the presence of heart disease. Their outputs are interpreted using techniques like SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-Agnostic Explanations). This research advances healthcare by bridging AI with human understanding, enhancing outcomes in heart disease management.

Problem Statement:

Heart disease is a major cause of mortality worldwide, making early detection critical for effective treatment. Machine learning models can predict heart disease with high accuracy, but their "black-box" nature limits adoption in clinical practice, as doctors often cannot understand the reasoning behind predictions. This lack of transparency reduces trust and hinders real-world implementation. There is a need for an approach that combines accuracy with interpretability. By integrating Explainable AI (XAI), predictive models can provide clear, feature-based explanations, enabling doctors to make informed decisions and improving patient outcomes.


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25/9/25
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